

HUDM 5001 Section 001 Programming for Data Science

Syllabus, Fall 2026

Version: 08.31.2026

Lecture Times: Tues 11:00 am - 12:40 pm, in-person

Location: Grace Dodge Hall (GDH) 449

Note: All times mentioned throughout the Canvas site and Syllabus are Eastern Time (US). Syllabus may be subject to change. Any revisions will be announced on Canvas and reflected in the version date above.

Instructor: [Chenguang Pan](#), cp3280@tc.columbia.edu

Office Hours: Tue. 3:00-4:00 pm. Online via Zoom:
<https://teacherscollege.zoom.us/j/92658353022>

When emailing me, please remember to include “HUDM5001” in your email subject line.

Course Overview and Learning Outcomes:

This course is an introduction to essential programming concepts, structures, and techniques for data science. Topics covered include data types, data structures, control statements, and functions, using the NumPy and Pandas libraries in the programming language Python. The course also covers version control using GitHub and database management using SQLite. Additionally, the course includes content on data visualizations and coding practices using AI.

At the end of the course, students will

- (1) Be able to confidently work in an appropriate programming environment (IDE).
- (2) Correctly describe basic Python language constructs and develop Python codes and write basic programs.
- (3) Understand version control concepts and work on a data science project using GitHub and Python.
- (4) Create a portfolio showcasing your visualization skills.

Textbooks

- **Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython**, 2nd Edition, McKinney. O'Reilly Media / ISBN: 978-1-4919-5766-0 Freely available through the library: <https://learning.oreilly.com/library/view/python-for-data/9781491957653/>

Software or Development Tools:

- [Anaconda](#)
- [Colab](#) (or other IDE tools like PyCharm, VSCode, etc.)
- [GitHub](#)
- [SQLite](#)

Topics Covered

- Python Programming
 - Intro to Colab, Conda, JupyterNotebook
 - variables and expressions
 - data types: int, float, bool, string, list, tuple, set, dict, range
 - operators
 - input/output
 - numpy
 - pandas
 - sqlite database
 - control structures
 - iterables and iterators
 - list comprehensions
 - functions
 - lambda functions
 - running scripts at the command line
 - classes
 - unit testing
- Python Data Visualization
 - Matplotlib, Plotnine, Plotly
- Intro to GitHub
- Intro to SQL
- Coding with AI
- Advanced topics (TBD)

Schedule of Topics and Assignments

A = Programming Assignments

Week	Date	Topics	Assignments	Reading
Week 1	9/8	Syllabus and Intro to Python	Pre-class Survey	

Week	Date	Topics	Assignments	Reading
Week 2	9/15	Python Intro: Data Types, Variables and Expressions	A	McKinney Chs 1&2; OM01
Week 3	9/22	Python Introduction: Operators, Input/Output, Numpy	A	McKinney Ch 4; OM02 ; OM03
Week 4	9/29	Python Intro: Pandas	A	McKinney Ch 5; OM04 ; OM05
Week 5	10/6	Pandas, SQLite Database	A	OM05 ; OM06
Week 6	10/13	Control Structures and Iterables	A	OM07 ; OM08
Week 7	10/20	Functions, Lambdas, List Comprehensions	A	McKinney Ch 3.2; OM09 ; OM10 ; OM11 ; OM12
Week 8	10/27	Midterm		
Week 9	11/3	Intro to Shell and GitHub. No in-person class due to College Holiday. Recording will be provided.	A	OM13 ; OM14 ; OM15
Week 10	11/10	Python Classes, Object-Oriented Programming	A	OM16 ; OM17

Week	Date	Topics	Assignments	Reading
Week 11	11/17	Unit Testing and Exception Handling	A	
Week 12	11/24	Advanced Topics (1): Coding Practices with AI		
Week 13	12/1	Advanced Topics (2): TBD		
Week 14	12/8	Jupyter Notebook Presentation		
Week 15	12/15	Final Project Discussion		
Week 16	12/22	Final Project	Final Report	

*OM means online materials. Coding Practices with AI will focus on vibe coding. Advanced Topics may cover one or two topics from Machine Learning, Deep Learning, Natural Language Processing, etc. It depends on teaching pace.

Programming Assignments, the Midterm, and the Final Project

Programming Assignments:

The programming assignments consist of focused exercises related to each week's lectures. You are expected to first try to complete the homework by yourself. If you work with others, please make sure that you understand all of the work, and that your final submission is your own work. The assignments will be posted on GitHub/Canvas no later than Tuesday and will be due the following Monday at 11:59pm, EST. The total possible points for each assignment may vary, and specific grading criteria will be provided with each assignment. I or our CA (if one is assigned) will view students' submissions and make comments on them. Assignments turned in late will be subject to the following penalty: 10% of the total score will be deducted for each day past the due date. An assignment with the lowest score will be dropped when computing the final letter grade at the end of the semester.

AI tools may be used on programming assignments under conditions described in the **AI Policy** section below. We will cover formatting details and any exceptions in our first class.

Midterm:

The midterm will cover Weeks 1–7, and will be an in-person, paper-based exam. You will be allowed one 8 ½ x 11 (the standard US letter used in printers) one-sided **cheat sheet**. You're allowed to bring a calculator.

AI tools are not allowed for the midterm.

Final Project:

You will work with other students in the same group. Pick a dataset that you and your group find interesting. Example sources are found below. Feel free to select your data from any other source as appropriate.

The final project should form a research question, and perform data pre-processing, data cleaning, outlier removal, and so on to sanitize your data as necessary. Explore your data to reveal interesting/useful information based on your project scenario, and create at least 2 visualizations that you find interesting/useful. Also, do at least one of the following, depending on your interests and background: (i) compute meaningful statistical quantities (e.g., means, correlations), (ii) perform a statistical test on the data (e.g., t-test), or (iii) fit a model to the data (e.g., regression).

The final report should cover the following sections: abstract, introduction, data, data processing methodology, results, and conclusions. Also, you should submit your Python codes, and make detailed annotations on the codes so that peers can easily reproduce your work. The files can be in Jupyter Notebooks or Python scripts. The maximum number of pages is limited to 10 pages (double spaced; excluding the appendix). The paper should be written as coherently as possible. More details about the final project will be announced on Canvas.

AI tools are not allowed for the final project.

Attendance:

Regular attendance is required for this course. Missing more than two sessions will have a negative impact on your final grade.

Data

For your final project, you will analyze real data and draw meaningful conclusions with regard to your research questions. Here is a list of websites where you can find interesting data.

- [kaggle](#)
- [AWS Open Data](#)
- [data.world](#)
- [ICPSR](#)
- [The Google Dataset Search](#)
- [The UCIML Repo](#)
- [The CMU data repository](#)
- [The datasets subreddit](#)
- [Tycho](#)
- [Data Portals](#)

Delivery Mode Expectations

- Students complete assigned reading before live sessions
- In-person live sessions will consist of:
 - the instructor gives code demos
 - students work on small and larger coding assignments, with assistance from instructor/TA/potentially their peers
 - the instructor reviews coding solutions with the class
 - students submit assignments through Canvas
- Note that this course is conducted in-person. In-person lectures will not be recorded using Zoom or any recording tools. However, in the event of emergencies such as COVID-19 or natural disasters, in-person lectures may be recorded via Zoom.

Electronic Submission of Assignments

All assignments must be submitted electronically through Canvas by the specified due dates and times. It is important to complete all assigned work—failure to do so will likely result in failing the class.

Class Management

Email / Communication

- Email is the best way to get in touch with the teaching staff: professor and CAs.
- Please be sure to include the course number ("HUDM5001") in your email subject line when sending an email to any of the teaching staff.

Grading

Courses at Teachers College use [the following grading system](#): A+, A, A-; B+, B, B-; C+, ..., F. The symbol W is used when a student officially drops a course before its completion or if the student withdraws from an academic program of the University.

Component	Weight
Programming Assignments (drop one)	40%
Midterm	20%
Presentation	10%
Final Project	30%

If your weighted total points are ...	Your final letter grade is ...
[93,100]	A
[90,93)	A-
[83,90)	B
[80,83)	B-
[73,80)	C
[70,73)	C-
<70	F

Note that A+, B+, and C+ will be determined by the class curve, attendance, and overall performance in the course.

AI Policy

Intellectual honesty is vital for an academic community and for the fair evaluation of your work by teaching staff. All work submitted in this course must be your own or that of your group, completed in accordance with the University's academic policies, and you may not engage in unauthorized collaboration.

Because working effectively with AI coding tools is part of what this course teaches, AI use is permitted on the weekly programming assignments under the conditions below. No AI tools are permitted on the midterm exam or the final project.

1. **Attempt it yourself first.** For each programming assignment, work through the question on your own before opening any AI tool, and save that unassisted attempt as your original draft.
2. **Then refine, if you want to.** You may use AI to debug, explain error messages, or improve your code. If you do, submit both the original draft and the revised draft, clearly labeled. An original draft that is incomplete or partly wrong is fine.
3. **Say what you used.** Include a brief note (a comment block at the top of your submission is enough) naming the tool and what you used it for.
4. **Own what you submit.** You are responsible for every line you turn in and should be able to explain how it works if asked.

Submitting an AI-generated solution in place of a genuine original draft will be treated as an academic-integrity issue under the Academic Integrity policy below.

Services for Students with Disabilities

The College will make reasonable accommodations for persons with documented disabilities. Students are encouraged to contact the Office of Access and Services for Individuals with Disabilities for information about registration (166 Thorndike Hall). Services are available only to students who are registered and submit appropriate documentation. As your instructor, I am happy to discuss specific needs with you as well.

IN Incomplete

The grade of Incomplete is to be assigned only when the course attendance requirement has been met but, for reasons satisfactory to the instructor, the granting of a final grade has been postponed because certain course assignments are outstanding. If the outstanding assignments are completed within one calendar year from the date of the close of term in which the grade of Incomplete was received and a final grade submitted, the final grade will be recorded on the permanent transcript, replacing the grade of Incomplete, with a transcript notation indicating the date that the grade of Incomplete was replaced by a final grade.

If the outstanding work is not completed within one calendar year from the date of the close of term in which the grade of Incomplete was received, the grade will remain as a permanent Incomplete on the transcript. In such instances, if the course is a required course or part of an approved program of study, students will be required to re-enroll in the course including repayment of all tuition and fee charges for the new registration and satisfactorily complete all

course requirements. If the required course is not offered in subsequent terms, the student should speak with the faculty advisor or Program Coordinator about their options for fulfilling the degree requirement. Doctoral students with six or more credits with grades of Incomplete included on their program of study will not be allowed to sit for the certification exam.

Email

Teachers College students have the responsibility for activating the Columbia University Network ID (UNI) and a free TC Gmail account. As official communications from the College – e.g., information on graduation, announcements of closing due to severe storm, flu epidemic, transportation disruption, etc. – will be sent to the student’s TC Gmail account, students are responsible for either reading email there, or, for utilizing the mail forwarding option to forward mail from their account to an email address which they will monitor.

Religious Holidays

It is the policy of Teachers College to respect its members’ observance of their major religious holidays. Students should notify instructors at the beginning of the semester about their wishes to observe holidays on days when class sessions are scheduled. Where academic scheduling conflicts prove unavoidable, no student will be penalized for absence due to religious reasons, and alternative means will be sought for satisfying the academic requirements involved. If a suitable arrangement cannot be worked out between the student and the instructor, students and instructors should consult the appropriate department chair or director. If an additional appeal is needed, it may be taken to the Provost.

Academic Integrity

Students who intentionally submit work either not their own or without clear attribution to the original source, fabricate data or other information, engage in cheating, or misrepresentation of academic records may be subject to charges. Sanctions may include dismissal from the college for violation of the TC principles of academic and professional integrity fundamental to the purpose of the College.

Sexual Harassment and Violence Reporting

Teachers College is committed to maintaining a safe environment for students. Because of this commitment and because of federal and state regulations, we must advise you that if you tell any of your instructors about sexual harassment or gender-based misconduct involving a member of the campus community, your instructor is required to report this information to the Title IX Coordinator, Janice Robinson. She will treat this information as private, but will need to follow up with you and possibly look into the matter. The Ombuds officer for Gender-Based Misconduct is a confidential resource available for students, staff and faculty. “Gender-based

misconduct” includes sexual assault, stalking, sexual harassment, dating violence, domestic violence, sexual exploitation, and gender-based harassment. For more information, see <http://sexualrespect.columbia.edu/gender-based-misconduct-policy-students>

Emergency Plan

TC is prepared for a wide range of emergencies. After declaring an emergency situation, the President/Provost will provide the community with critical information on procedures and available assistance. If travel to campus is not feasible, instructors will facilitate academic continuity through Canvas and other technologies, if possible.

1. It is the student’s responsibility to ensure that they are set to receive email notifications from TC and communications from their instructor at their TC email address.
2. Within the first two sessions for the course, students are expected to review and be prepared to follow the instructions stated in the emergency plan.
3. The plan may consist of downloading or obtaining all available readings for the course or the instructor may provide other instructions.